

Body Mass Index and Migraine Burden: A Literature Review

• Introduction:

Migraine is a common primary headache disorder characterized by recurrent attacks of moderate to severe headache that are frequently associated with nausea, vomiting, photophobia, and phonophobia. It is one of the leading causes of disability worldwide, particularly among individuals under the age of 50 years, and has a significant impact on quality of life, daily functioning, and healthcare costs. "Migraine is common, and occurs in about 12% of the US population overall, 18% of women, and 6% of men, each year". (Burch et al., 2019).

Migraine has both non-modifiable risk factors including female sex, genetic predisposition, and family history. And modifiable risk factors including stress, sleep disturbances, dietary habits, physical inactivity, and obesity.

Among these, obesity is an important risk factor for migraine.

How we can define obesity ?

Obesity means having too much fat tissue. The most common way to measure body fat is body mass index, referred as BMI.

BMI categories a person's weight as underweight, healthy weight, overweight, or obese. We can calculate the BMI by using the Height and Weight and applying a Formula:

$(\text{Weight}(\text{in pounds}) / \text{height}(\text{in inches})^2) * 703$

The WHO defines obesity as having a BMI of at least 30.

Both conditions are highly prevalent worldwide and contribute to a significant burden on individuals and healthcare systems, that is mean why both are important

This literature review focuses on the association between obesity and migraine in adults. It explain the increase in BMI on migraine frequency, and severity. It does not focus on other types of headaches or secondary causes. Specifically, it seeks to answer the following research question :

Among adults, is obesity (high BMI) associated with increased frequency and severity of migraine compared to individuals with normal BMI?

• Theoretical framework:

What is the obesity according to WHO intro ?

"Overweight and obesity are defined as abnormal or excessive fat accumulation that presents a risk to health. A body mass index (BMI) over 25 is considered overweight, and over 30 is obese. In 2019, an estimated 5 million noncommunicable disease (NCD) deaths were caused by higher-than-optimal BMI." WHO intro.

So the WHO BMI categories were:

1. Underweight :
BMI < 18.5 kg/m²
2. Normal weight :
BMI = 18.5 – 24.9 kg/m²
3. Pre-obesity / Overweight :
BMI = 25.0 – 29.9 kg/m²
4. Obesity :
BMI ≥ 30.0 kg/m²

What is the migraine?

The migraine is not just a bad headache. It's a disabling neurological disease with different Symptom and different treatment compared with other headaches disorders.

Migraine is classified into several subtypes. The major types include:

1. Migraine without aura
2. Migraine with aura
3. Chronic migraine

- Diagnostic Criteria for Migraine without Aura (ICHD-3)

- A. At least five attacks fulfilling criteria B–D
- B. Headache attacks lasting 4–72 hours (untreated or unsuccessfully treated)
- C. Headache has at least two of the following characteristics:
 1. Unilateral location
 2. Pulsating quality
 3. Moderate or severe pain intensity
 4. Aggravation by or causing avoidance of routine physical activity (e.g., walking or climbing stairs)
- D. During headache, at least one of the following:
 1. Nausea and/or vomiting
 2. Photophobia and phonophobia
- E. Not better accounted for by another ICHD-3 diagnosis

- Diagnostic Criteria for Migraine with Aura (ICHD-3)

- A. At least two attacks fulfilling criteria B and C
- B. One or more of the following fully reversible aura symptoms:
 1. visual
 2. sensory
 3. speech and/or language
 4. motor
 5. brainstem
 6. retinal
- C. At least three of the following six characteristics:
 1. at least one aura symptom spreads gradually over ≥ 5 minutes
 2. two or more aura symptoms occur in succession
 3. each individual aura symptom lasts 5–60 minutes¹
 4. at least one aura symptom is unilateral²
 5. at least one aura symptom is positive³
 6. the aura is accompanied, or followed within 60 minutes, by headache
- D. Not better accounted for by another ICHD-3 diagnosis.

The pathophysiology of migraine involves a combination of neuronal, vascular, and inflammatory mechanisms. Activation of the trigeminovascular system leads to the release of vasoactive neuropeptides, particularly calcitonin gene-related peptide (CGRP), substance P, and neurokinin A. These mediators cause vasodilation of cerebral and meningeal blood vessels, neurogenic inflammation, and sensitization of pain pathways. Repeated activation of pain pathways leads to the persistence and increased frequency of migraine attacks.

What is the link between obesity and migraine?

Since 1990, experts have come to believe that fat is highly active substance. In people who are obese, the extra fat cells tell the body to make inflammatory proteins. This new understanding of fat cells suggests that obesity keeps the body in a mild but constant state of inflammation. As part of this inflammatory state, the body produces pain-generating hormones and releases them from fat cells. During a migraine attack, there is a similar release of these same pain-promoting hormones. This means that obese individuals may have an extra factor that increases pain levels and headache frequency. (American Migraine Foundation. (n.d.)).

- **Review of previous studies:**

1. Migraine and Obesity: Is there a relationship?

Several studies have reported a positive association between obesity and migraine. Ford et al.(2008) conducted a population-based study in the United States to investigate the relationship between body mass index (BMI) and headache disorders. The study found that obesity was associated with a higher prevalence of migraine compared with normal-weight individuals. One of the main strengths of this study was the use of a large national sample, which increased the reliability of the findings. However, because the study had a cross-sectional design, it was unable to establish a causal relationship between obesity and migraine.

In contrast, Mattsson (2007) conducted a population-based study in Sweden involving women aged 40-74 years and found no significant association between obesity and migraine. The study benefited from a well-defined study population and standardized assessment methods. Nevertheless, its findings may not be generalizable because only middle-aged and older women were included, and younger adults and men were not represented.

2. Calcitonin gene-related peptide: A molecular link between obesity and migraine?

Recober and Goadsby (2010) propose that calcitonin gene-related peptide (CGRP) may serve as a biological link between obesity and migraine. They suggest that obesity-related inflammatory mediators may increase CGRP activity, thereby lowering the threshold for migraine attacks and potentially contributing to chronic migraine progression.

Similarly, Maasumi et al. (2018) explained that CGRP plays a central role in the pathophysiology of migraine and represents a key therapeutic target. Their review focused on the therapeutic effects of blocking CGRP ligand and receptor, providing strong evidence for the importance of CGRP in migraine mechanisms. The findings support the hypothesis that CGRP-related pathways may be influenced by metabolic and inflammatory conditions.

Furthermore, Edvinsson (2017) highlighted the central role of calcitonin gene-related peptide (CGRP) in migraine pathophysiology through neurogenic inflammation and trigeminovascular activation, emphasizing its importance as a therapeutic target. However, the review primarily describes CGRP within neurological pathways rather than metabolic conditions such as obesity.

3. Migraine and Body Mass Index categories:

Several studies have investigated the association between obesity and migraine, but their findings have been inconsistent. Bigal et al. (2006) conducted a large population-based cross-sectional study in the United States to examine the relationship between body mass index (BMI) and migraine. The study reported that obese individuals had a higher prevalence of migraine than those with normal body weight, suggesting that obesity may be a risk factor for migraine. A major strength of this study was its large sample size and population-based design; however, its cross-sectional nature limited the ability to establish a causal relationship between obesity and migraine.

Similarly, Peterlin et al. (2013) carried out a cross-sectional study in the United States to investigate whether age, sex, and race influenced the relationship between obesity and episodic migraine. The authors found that obesity was associated with an increased risk of migraine, particularly among women and younger adults. This study was valuable because it considered important modifying factors such as age and sex. Nevertheless, because it was observational, it could not determine whether obesity directly caused migraine.

Le et al. (2011) performed a population-based cross-sectional study in Denmark and reported different findings. Although the study did not demonstrate an overall increase in migraine risk among obese individuals, it found a decreased risk of migraine in men in some BMI categories. The researchers also evaluated lifestyle and socioeconomic factors that could influence the association, which strengthened the study. However, the cross-sectional design and the possibility of residual confounding limited the interpretation of the results.

4. The Association Between Migraine and Body Mass Index (BMI):

The 2025 study aimed to evaluate the association between BMI, waist circumference, and the risk of developing migraine. It was designed as a large prospective cohort study, meaning that researchers followed a large group of adults over several years to observe the occurrence of migraine while recording participants' body weight, height, and other health-related factors. This study design is considered reliable because it allows researchers to examine how changes in body weight influence future health outcomes.

The researchers categorized participants into different BMI groups, including underweight, normal weight, overweight, and obese. They also adjusted for several possible confounding factors such as age, sex, smoking status, physical activity, and existing medical conditions to ensure that the observed association was primarily related to BMI.

Main Findings:

The study demonstrated that individuals with obesity had a significantly higher risk of developing migraine compared with those who maintained a normal BMI. Furthermore, the risk increased progressively as BMI increased, suggesting a dose-response relationship between excess body weight and migraine. Participants with central obesity, measured by waist circumference, also showed a higher likelihood of experiencing migraine attacks. The researchers proposed several biological mechanisms to explain this association. Excess body fat promotes chronic low-grade inflammation by increasing inflammatory cytokines such as tumor necrosis factor-alpha (TNF- α) and interleukin-6 (IL-6). These inflammatory mediators may increase the sensitivity of the trigeminal pain pathway.

Clinical Significance:

Lifestyle interventions such as weight reduction, regular physical exercise, a balanced diet, and improved sleep habits may decrease both the frequency and severity of migraine attacks. The study also emphasizes the importance of early identification of overweight and obese individuals who experience recurrent headaches. Counseling patients about maintaining a healthy BMI may improve long-term neurological outcomes and reduce healthcare costs associated with chronic migraine.

Limitations:

Despite its strengths, the study has some limitations. Since it was observational, it could identify an association but could not definitively prove that obesity causes migraine. In addition, BMI does not distinguish between fat mass and muscle mass, meaning it may not perfectly reflect body composition. Future research should investigate whether intentional weight loss directly reduces migraine frequency through randomized controlled clinical trials.

- **Critical analysis:**

Overall, the available evidence suggests a positive association between obesity and migraine; however, the strength and consistency of this relationship vary across studies. Most epidemiological studies, including Ford et al. (2008), Bigal et al. (2006), Peterlin et al. (2013), and the 2025 prospective cohort study, support an association between higher BMI and increased prevalence or risk of migraine. The 2025 cohort study provides stronger evidence compared to earlier cross-sectional studies due to its longitudinal design, which allows assessment of temporal relationships between obesity and the development of migraine.

Despite these findings, important methodological limitations exist. The majority of earlier studies are cross-sectional in nature, which limits causal inference and only allows identification of associations rather than directionality. In addition, reliance on BMI as the primary measure of obesity may reduce accuracy, as it does not distinguish between fat mass and lean body mass or account for fat distribution. Residual confounding factors such as diet, socioeconomic status, physical activity, and comorbid conditions may also influence the observed associations.

The findings across studies are not entirely consistent. While most studies report a positive association, Mattsson (2007) found no significant relationship between obesity and migraine in a specific population of middle-aged and older women, and Le et al. (2011) reported heterogeneous associations depending on sex and BMI categories. These discrepancies may be explained by differences in study design, population characteristics, diagnostic criteria for migraine, and variability in BMI classification.

From a mechanistic perspective, CGRP-related pathways have been proposed as a potential biological link between obesity and migraine. Recober and Goadsby (2010) suggested that inflammatory mediators associated with obesity may influence CGRP activity, while Maasumi et al. (2018) and Edvinsson (2017) highlighted the central role of CGRP in migraine pathophysiology. However, despite these hypotheses, direct mechanistic evidence linking obesity-induced metabolic changes to CGRP activation remains limited, indicating that the biological pathway is not yet fully established.

Overall, although current evidence supports an association between obesity and migraine, inconsistencies across studies and limited mechanistic understanding mean that the causal relationship remains to be fully established.

- **Research Gap:**

Although current evidence suggests an association between obesity and migraine, important gaps remain in the literature. Most studies are observational and cross-sectional, limiting the ability to establish causality and the temporal relationship between BMI and migraine development.

There is also inconsistency in findings across populations, suggesting that factors such as age, sex, and lifestyle may modify the association, but these interactions are not yet fully understood. Furthermore, although inflammatory pathways and CGRP have been proposed as potential biological links, the exact mechanisms underlying the relationship between obesity and migraine remain unclear.

Future research should focus on large prospective cohort studies and interventional trials, particularly those examining whether weight reduction can reduce migraine frequency and severity.

- **Conclusion:**

The current literature suggests that obesity is associated with an increased risk and prevalence of migraine, although the strength of this association varies across studies. Most evidence supports a positive relationship, particularly among women and younger adults, while a few studies have reported inconsistent findings due to differences in study populations, research designs, and methodological approaches. Current evidence also suggests that inflammatory pathways and calcitonin gene-related peptide (CGRP) may contribute to the biological link between obesity and migraine; however, these mechanisms have not yet been fully established. Overall, the available research indicates that obesity is an important modifiable risk factor that may influence migraine occurrence and severity.

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